

Please note: this “submission” was created by ChatGPT as an example. This submission was created following the process given in the process document.

Example Mini-Capstone Project Report

Using Generative AI to Support Personalized Learning in Experimental Methods

For this project, I explored how generative AI could support personalized learning in an aerospace engineering Experimental Methods course. Students often struggle with core instrumentation concepts, but not for the same reasons. Some students want equations first, others want pictures, others need procedures spelled out step-by-step, and others want a big-picture intuitive explanation. My goal was to see whether generative AI could help create adaptive explanations tailored to different learning preferences.

I chose this task because personalized learning aligns well with generative AI’s strengths: it can adjust style, structure, tone, and level of detail instantly. I started with a broad idea of “AI for personalized experimental methods learning,” but I asked a chatbot to help me refine the scope. The model suggested focusing on **one topic** and building a miniature prototype that adapts explanations to different learner types. I selected **Pitot-static pressure measurement**—a core topic in subsonic wind-tunnel testing—because it is visual, mathematical, and widely used in both labs and real aircraft.

To explore the idea, I used two generative AI tools. My first tool was a text-based chatbot. I asked it to explain Pitot-static pressure measurement in four different styles: visual, analytical, step-by-step procedural, and big-picture conceptual. The results were better than I expected. Each explanation felt distinct and aligned with the intended learning preference. The analytical version walked through Bernoulli’s equation and pressure relationships, while the procedural version listed concrete steps for using the instrumentation. The visual and big-picture versions relied more on imagery and real-world context. This output convinced me that text-based AI can be an effective partner for generating personalized explanations quickly.

My second tool was a multimodal image generator accessed through LM Arena. I asked it to create a labeled diagram of a Pitot-static tube. The images it produced were clean, high-resolution, and aesthetically polished—artistically, they were excellent. However, they were not **technically usable**. The geometry, dimensions, and port placements were inaccurate, and some visual elements looked stylized rather than realistic. This mismatch

highlighted a key limitation: multimodal models can produce beautiful images that are wrong in ways that matter for engineering instruction.

Reflecting on the process, I was surprised by how helpful the text model was at generating useful ideas and explanation variants. The chatbot expanded the range of instructional approaches I could offer and did so with very little prompting. On the other hand, the image model reinforced how important accuracy is in technical fields. Even small visual errors could mislead students or contradict physical reality. The contrast made it clear that different generative tools have different strengths, and that selecting the right tool for the task is essential.

There were also ethical and practical considerations. The inaccurate diagrams show how generative AI can unintentionally spread misinformation, especially when outputs appear polished and authoritative. Students might trust an incorrect diagram simply because it “looks official.” On the text side, AI can gloss over caveats, oversimplify assumptions, or generalize in ways that aren’t always appropriate for a specific lab setup. Transparency is also important—if AI-generated content is used in a course, learners should know where it came from and understand its limitations. Finally, if AI systems rely on uploaded documents, issues of data ownership, privacy, and responsible handling must be addressed.

In terms of outcomes, AI absolutely helped me move toward my goal. It accelerated the process of generating learning-style-specific explanations and pushed me to think more creatively about instructional design. However, I also learned that these outputs cannot be used without technical review. I would trust a chatbot to help with brainstorming and early drafts, but not with final instructional materials. I would not rely on multimodal tools for engineering diagrams unless the images were corrected by hand.

Ultimately, this project helped me better understand both what AI can do well and where it fails. I learned that AI is best used as a collaborator that expands possibilities, not as an authoritative source. I also learned that I need to approach AI-generated content critically—questioning assumptions, verifying details, and adapting outputs to ensure technical correctness. Generative AI can enhance personalized learning, but only when paired with human judgment, expertise, and responsibility.